

AI Deployment, Security & Compliance – Scenario-Based Quiz

Instructions

- Choose the **best answer (A/B/C/D)**
 - Questions are based on **real-world situations**
 - No direct theory recall — apply concepts
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Section 1: AI Deployment Strategies

Q1

A startup deploys an AI chatbot directly on users' devices instead of the cloud to reduce latency and protect sensitive data.

What deployment strategy is being used?

- A. Centralized cloud deployment
 - B. Edge deployment**
 - C. Hybrid deployment
 - D. Multi-cloud deployment
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Q2

A company runs its AI model in Azure but stores backup models in AWS for reliability. Which strategy is this?

- A. Edge computing
 - B. Multi-cloud deployment**
 - C. On-premise deployment
 - D. Serverless deployment
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Q3

An AI fraud detection system must respond within milliseconds and cannot rely on internet connectivity.

Which deployment is best?

- A. Cloud deployment
 - B. Edge deployment**
 - C. Hybrid deployment
 - D. Batch processing
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Q4

A healthcare company processes patient data locally but uses cloud AI models for predictions. What type of deployment is this?

- A. Hybrid deployment**
 - B. Edge-only deployment
 - C. Centralized deployment
 - D. Offline deployment
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Q5

An AI model is deployed using containers and automatically scales based on traffic demand. Which concept is applied?

- A. Manual scaling
 - B. Serverless architecture
 - C. Containerized deployment with auto-scaling**
 - D. Static infrastructure
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Section 2: Identity & Access Management (IAM)

Q6

A company ensures that developers can only access AI model logs but not modify the model. Which principle is being followed?

- A. Full access control
 - B. Least privilege**
 - C. Open access
 - D. Shared responsibility
-

Q7

An AI system requires users to log in using both password and OTP.
What security method is this?

- A. Single sign-on
 - B. Multi-factor authentication**
 - C. Role-based access
 - D. Encryption
-

Q8

A cloud admin assigns permissions based on roles like "Data Scientist" and "Viewer".
What model is this?

- A. Attribute-based access control
 - B. Role-based access control (RBAC)**
 - C. Mandatory access control
 - D. Open access control
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Q9

An employee leaves the company, but their AI system access is not revoked.
What risk does this create?

- A. Performance issue
 - B. Security vulnerability**
 - C. Data redundancy
 - D. Scalability issue
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Q10

A system allows temporary access tokens instead of permanent credentials.
Why is this safer?

- A. Faster performance
 - B. Reduced risk of credential misuse**
 - C. Easier implementation
 - D. Lower cost
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Section 3: Responsible AI Principles

Q11

An AI hiring system favors male candidates over female candidates due to biased training data.
Which issue is this?

- A. Overfitting
 - B. Bias in AI**
 - C. Underfitting
 - D. Data leakage
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Q12

A bank wants to ensure its AI loan approval system explains why a loan was rejected.
Which principle is applied?

- A. Privacy
 - B. Transparency**
 - C. Scalability
 - D. Efficiency
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Q13

An AI system is tested across different demographic groups before deployment.
What principle is being ensured?

- A. Security
- B. Fairness**

- C. Latency
 - D. Automation
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Q14

An organization logs all AI decisions for audit purposes.
Which principle does this support?

- A. Accountability
 - B. Scalability
 - C. Optimization
 - D. Redundancy
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Q15

An AI model avoids using sensitive attributes like religion or caste.
What principle is followed?

- A. Bias amplification
 - B. Ethical AI
 - C. Data compression
 - D. Feature engineering
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Section 4: Security, Privacy & Compliance

Q16

A company encrypts user data before storing it in the database.
What is this protecting?

- A. Data integrity
 - B. Data confidentiality
 - C. Data availability
 - D. Data redundancy
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Q17

An AI system masks personal data like phone numbers before processing.
What technique is this?

- A. Encryption
 - B. Data anonymization**
 - C. Data replication
 - D. Data compression
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Q18

A company follows GDPR rules while handling European user data.
What does this ensure?

- A. Performance
 - B. Compliance**
 - C. Scalability
 - D. Automation
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Q19

An attacker intercepts data between user and server.
Which solution prevents this?

- A. Firewalls
 - B. Encryption (HTTPS/TLS)**
 - C. Load balancing
 - D. Caching
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Q20

A company regularly audits its AI systems for security vulnerabilities.
What practice is this?

- A. Debugging
- B. Compliance monitoring**
- C. Data cleaning
- D. Model tuning



Section 5: Mixed Real-World Scenarios

Q21

An AI healthcare system stores sensitive patient data and must follow strict regulations. What is the top priority?

- A. Speed
 - B. Security & compliance
 - C. UI design
 - D. Cost reduction
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Q22

An AI app crashes when too many users access it at once. Which deployment improvement is needed?

- A. Data cleaning
 - B. Auto-scaling
 - C. Encryption
 - D. Authentication
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Q23

A company uses logs to track who accessed AI predictions and when. What does this ensure?

- A. Transparency
 - B. Accountability
 - C. Efficiency
 - D. Redundancy
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Q24

An AI system is trained on incomplete data and performs poorly in production.
What issue occurred?

- A. Deployment failure
 - B. Poor data quality
 - C. Security breach
 - D. Compliance violation
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Q25

A company uses both IAM policies and encryption together.
What does this combination provide?

- A. Faster processing
- B. End-to-end security
- C. Reduced storage
- D. Simpler deployment